

Composite

The screenshot displays the IntelliJ IDEA IDE interface. On the left, the Project Explorer shows a project named 'Tree_structure' with a package structure: 'org.example' containing 'adapter', 'composite', and 'observer' packages. The 'Test_Composite' class is selected in the 'composite' package. The main editor shows the code for 'Test_Composite.java'.

```
1 package org.example.composite;
2
3 public class Test_Composite {
4     public static void main(String[] args) {
5
6         Folder root = new Folder( name: "Root");
7
8         File file1 = new File( name: "file1.txt");
9         File file2 = new File( name: "file2.txt");
10
11         Folder subFolder = new Folder( name: "Documents");
12
13         subFolder.add(new File( name: "doc1.pdf"));
14         subFolder.add(new File( name: "doc2.pdf"));
15
16         root.add(file1);
17         root.add(file2);
18         root.add(subFolder);
19
20         root.display();
21     }
22 }
```

Below the editor, the Run tab shows the execution command and output:

```
C:\Users\nguye\.jdk\ms-21.0.8\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2024.3.1\lib\idea_rt.jar=62613:C:\Program Files\JetBrains\IntelliJ IDEA 2024.3.1\bin" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath T:\KTMT\Tree_structure\target\classes org.example.composite.Test_Composite
```

Output:

```
Folder: Root
File: file1.txt
File: file2.txt
Folder: Documents
File: doc1.pdf
File: doc2.pdf
```

Observer

```
1 package org.example.observer;
2
3 public class Test_Observer {
4
5     public static void main(String[] args) {
6
7         Stock stock = new Stock();
8
9         Investor inv1 = new Investor(name: "Alice");
10        Investor inv2 = new Investor(name: "Bob");
11
12        stock.attach(inv1);
13        stock.attach(inv2);
14
15        stock.setPrice(100);
16        stock.setPrice(120);
17    }
18
19
20
```

Run Test_Observer

```
C:\Users\nguye\.jdk\ms-21.0.8\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2024.3.1\lib\idea_rt.jar=62622:C:\Program Files\JetBrains\IntelliJ IDEA 2024.3.1.1\bin" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath T:\KTMT\Tree_structure\target\classes org.example.observer.Test_Observer
Alice received stock price update: 100.0
Bob received stock price update: 100.0
Alice received stock price update: 120.0
Bob received stock price update: 120.0
```

Adapter

```
1 package org.example.adapter;
2
3 public class Test_Adapter {
4     public static void main(String[] args) {
5
6         XMLService xmlService = new XMLService();
7
8         JSONService adapter = new JSONAdapter(xmlService);
9
10        adapter.sendJSON("{name:'John'}");
11    }
12
13
14
```

Run Test_Adapter

```
C:\Users\nguye\.jdk\ms-21.0.8\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2024.3.1\lib\idea_rt.jar=51755:C:\Program Files\JetBrains\IntelliJ IDEA 2024.3.1.1\bin" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath T:\KTMT\Tree_structure\target\classes org.example.adapter.Test_Adapter
Sending XML: <data>{name:'John'}</data>
```